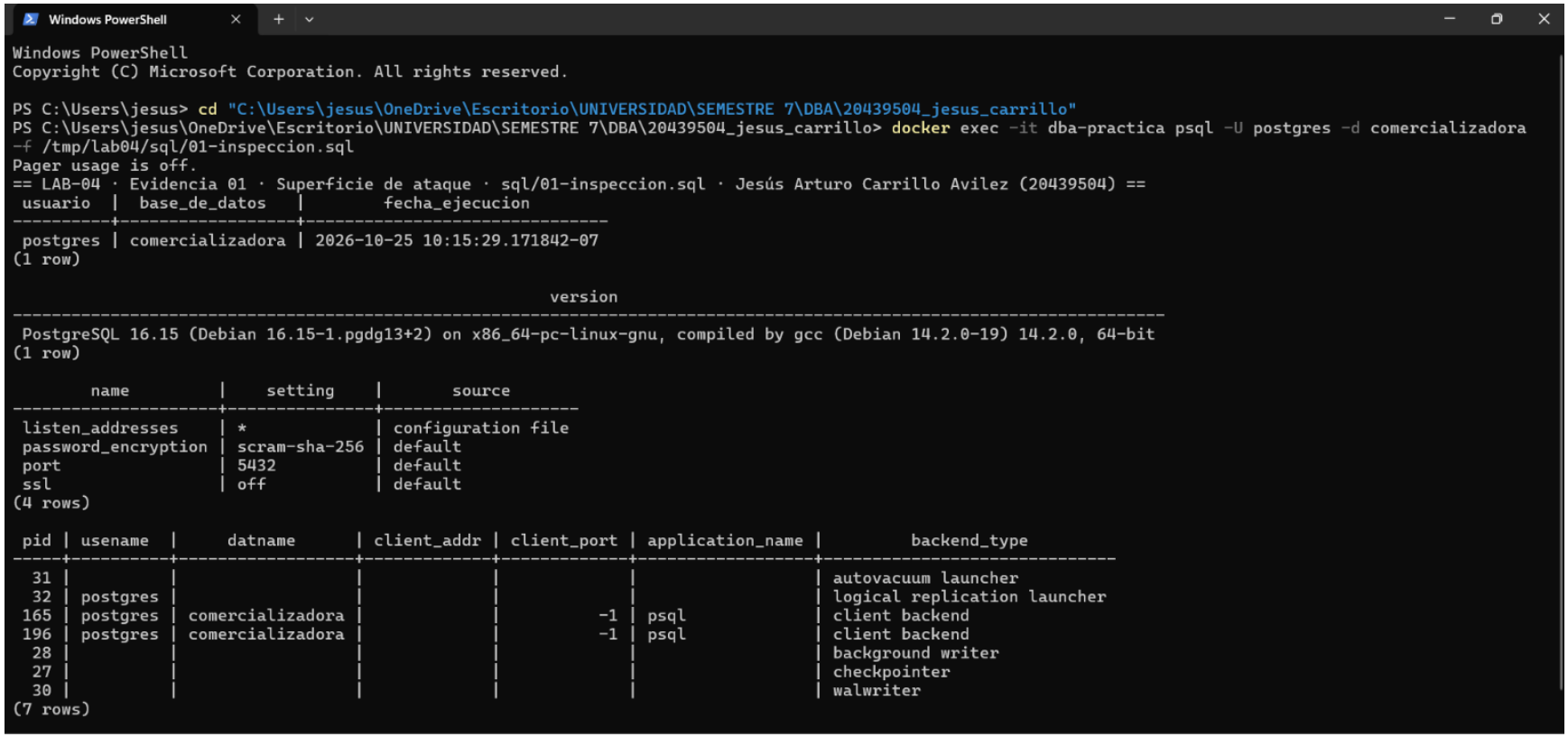
LAB-04 — Hardening y configuración segura de PostgreSQL

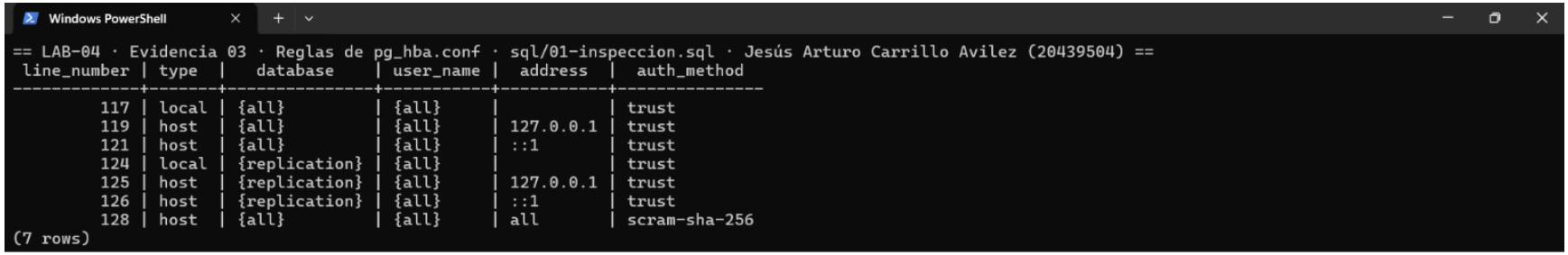
Evidencias 01 a 07 (sección 27 de la guía)

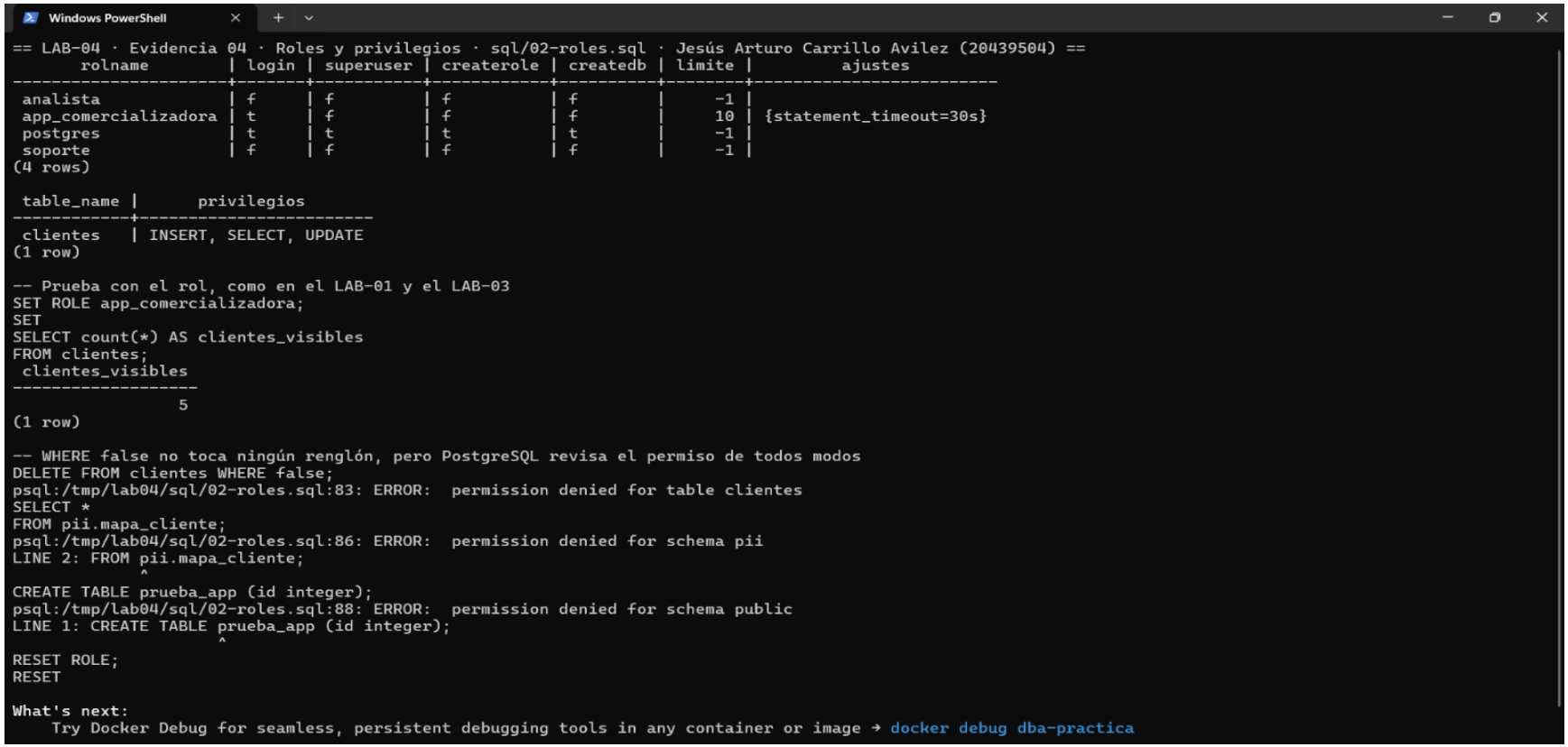
Gestión de Seguridad y Desempeño de Bases de Datos · Unidad II, tema 2.6

Jesús Arturo Carrillo Avilez · Cuenta 20439504 · Grupo 4 · Periodo 2026-2027-1

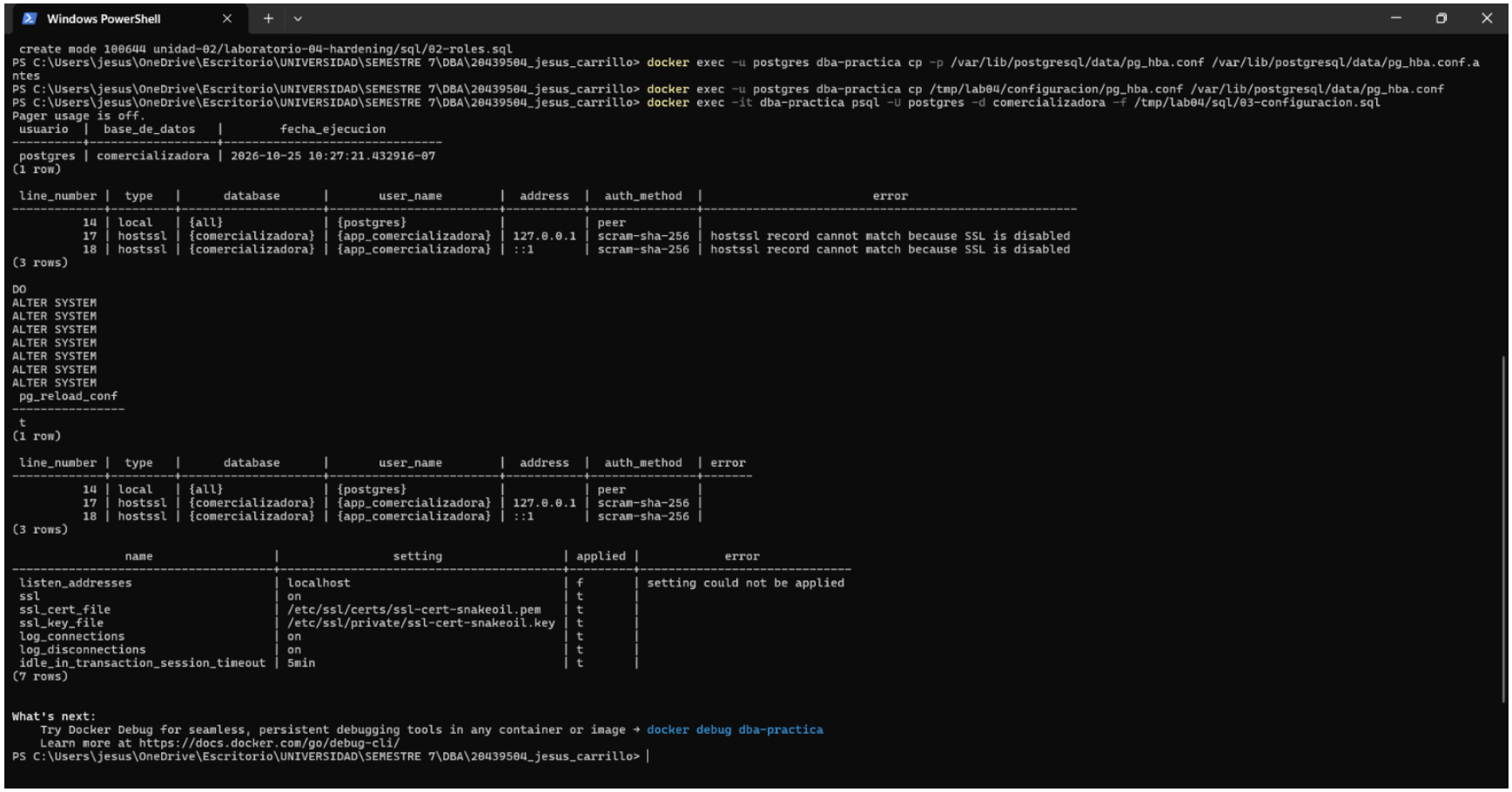
25 de octubre de 2026

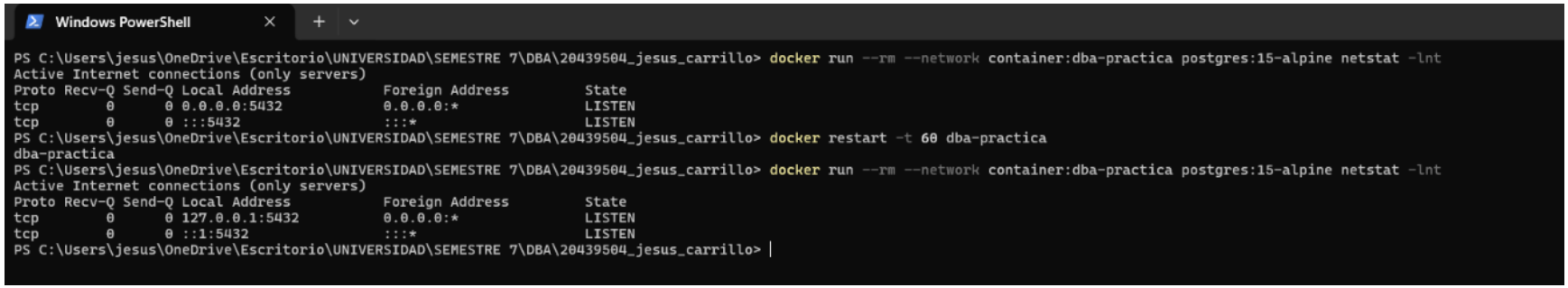


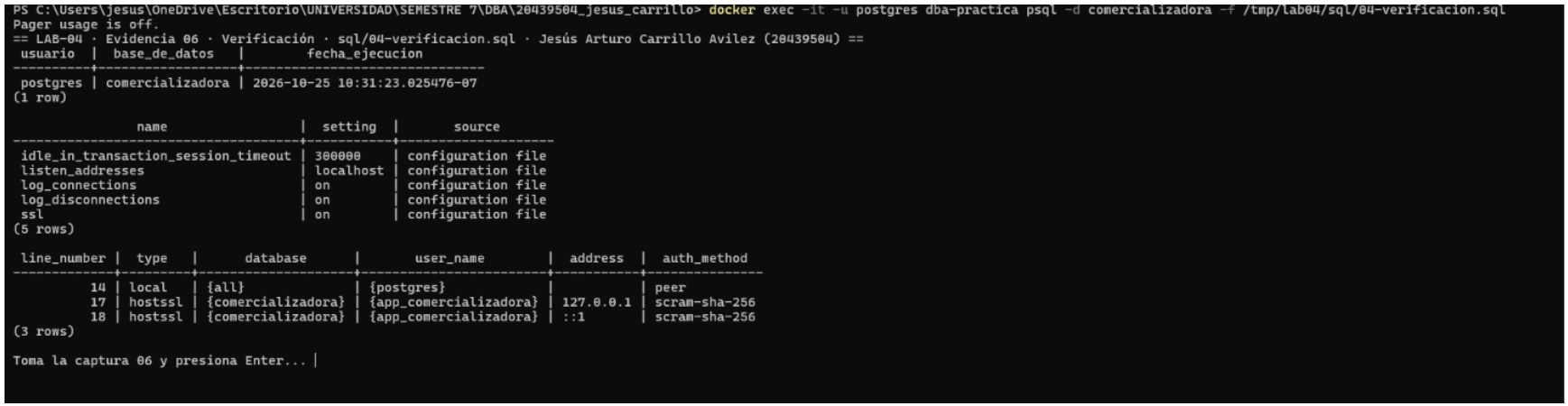

Windows PowerShell
== LAB-04 · Evidencia 03 · Reglas de pg_hba.conf · sql/01-inspeccion.sql · Jesús Arturo Carrillo Avilez (20439504) ==
 line_number | type | database | user_name | address | auth_method
-------------+-------+---------------+-----------+-----------+---------------
         117 | local | {all} | {all} | | trust
         119 | host | {all} | {all} | 127.0.0.1 | trust
         121 | host | {all} | {all} | ::1 | trust
         124 | local | {replication} | {all} | | trust
         125 | host | {replication} | {all} | 127.0.0.1 | trust
         126 | host | {replication} | {all} | ::1 | trust
         128 | host | {all} | {all} | all | scram-sha-256
(7 rows)

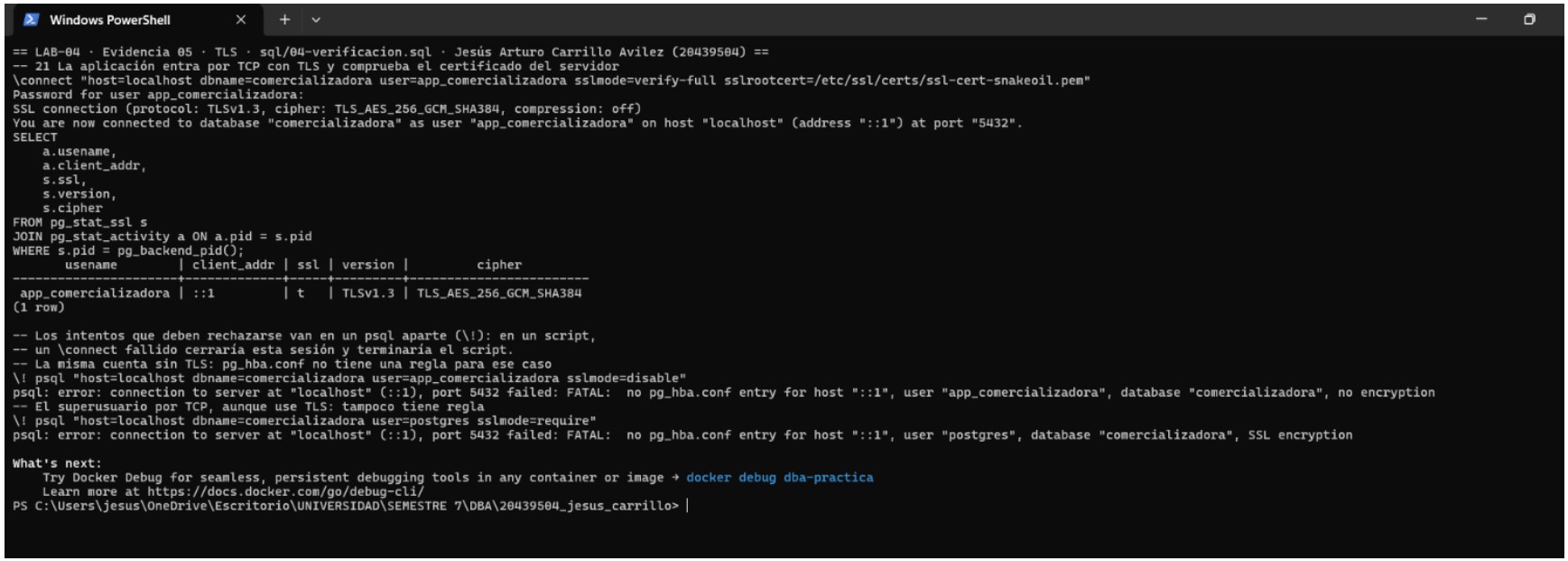

Windows PowerShell
== LAB-04 · Evidencia 04 · Roles y privilegios · sql/02-roles.sql · Jesús Arturo Carrillo Avilez (20439504) ==
       rolname | login | superuser | createrole | createdb | limite | ajustes
----------------------+-------+-----------+------------+----------+--------+-------------------------
 analista | f | f | f | f | -1 |
 app_comercializadora | t | f | f | f | 10 | {statement_timeout=30s}
 postgres | t | t | t | t | -1 |
 soporte | f | f | f | f | -1 |
(4 rows)
 table_name | privilegios
------------+------------------------
 clientes | INSERT, SELECT, UPDATE
(1 row)
-- Prueba con el rol, como en el LAB-01 y el LAB-03
SET ROLE app_comercializadora;
SET
SELECT count(*) AS clientes_visibles
FROM clientes;
 clientes_visibles

                 5
(1 row)
-- WHERE false no toca ningún renglón, pero PostgreSQL revisa el permiso de todos modos
DELETE FROM clientes WHERE false;
psql:/tmp/lab04/sql/02-roles.sql:83: ERROR: permission denied for table clientes
SELECT *
FROM pii.mapa_cliente;
psql:/tmp/lab04/sql/02-roles.sql:86: ERROR: permission denied for schema pii
LINE 2: FROM pii.mapa_cliente;
             ^
CREATE TABLE prueba_app (id integer);
psql:/tmp/lab04/sql/02-roles.sql:88: ERROR: permission denied for schema public
LINE 1: CREATE TABLE prueba_app (id integer);
                     ^
RESET ROLE;
RESET
What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug dba-practica


Windows PowerShell
create mode 100644 unidad-02/laboratorio-04-hardening/sql/02-roles.sql
PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo> docker exec -u postgres dba-practica cp -p /var/lib/postgresql/data/pg_hba.conf /var/lib/postgresql/data/pg_hba.conf.antes
PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo> docker exec -u postgres dba-practica cp /tmp/lab04/configuracion/pg_hba.conf /var/lib/postgresql/data/pg_hba.conf
PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo> docker exec -it dba-practica psql -U postgres -d comercializadora -f /tmp/lab04/sql/03-configuracion.sql
Pager usage is off.
usuario | base_de_datos | fecha_ejecucion
postgres | comercializadora | 2026-10-25 10:27:21.432916-07
(1 row)
line_number | type | database | user_name | address | auth_method | error
14 | local | {all} | {postgres} | | peer |
17 | hostssl | {comercializadora} | {app_comercializadora} | 127.0.0.1 | scram-sha-256 | hostssl record cannot match because SSL is disabled
18 | hostssl | {comercializadora} | {app_comercializadora} | ::1 | scram-sha-256 | hostssl record cannot match because SSL is disabled
(3 rows)
DO
ALTER SYSTEM
ALTER SYSTEM
ALTER SYSTEM
ALTER SYSTEM
ALTER SYSTEM
ALTER SYSTEM
ALTER SYSTEM
pg_reload_conf
t
(1 row)
line_number | type | database | user_name | address | auth_method | error
14 | local | {all} | {postgres} | | peer |
17 | hostssl | {comercializadora} | {app_comercializadora} | 127.0.0.1 | scram-sha-256 |
18 | hostssl | {comercializadora} | {app_comercializadora} | ::1 | scram-sha-256 |
(3 rows)
name | setting | applied | error
listen_addresses | localhost | f | setting could not be applied
ssl | on | t |
ssl_cert_file | /etc/ssl/certs/ssl-cert-snakeoil.pem | t |
ssl_key_file | /etc/ssl/private/ssl-cert-snakeoil.key | t |
log_connections | on | t |
log_disconnections | on | t |
idle_in_transaction_session_timeout | 5min | t |
(7 rows)
What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug dba-practica
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo>


Windows PowerShell
PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo> docker run --rm --network container:dba-practica postgres:15-alpine netstat -lnt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address Foreign Address State
tcp 0 0 0.0.0.0:5432 0.0.0.0:* LISTEN
tcp 0 0 :::5432 :::* LISTEN
PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo> docker restart -t 60 dba-practica
dba-practica
PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo> docker run --rm --network container:dba-practica postgres:15-alpine netstat -lnt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address Foreign Address State
tcp 0 0 127.0.0.1:5432 0.0.0.0:* LISTEN
tcp 0 0 ::1:5432 :::* LISTEN
PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo>


PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo> docker exec -it -u postgres dba-practica psql -d comercializadora -f /tmp/lab04/sql/04-verificacion.sql
Pager usage is off.
== LAB-04 · Evidencia 06 · Verificación · sql/04-verificacion.sql · Jesús Arturo Carrillo Avilez (20439504) ==
usuario | base_de_datos | fecha_ejecucion
postgres | comercializadora | 2026-10-25 10:31:23.025476-07
(1 row)
name | setting | source
idle_in_transaction_session_timeout | 300000 | configuration file
listen_addresses | localhost | configuration file
log_connections | on | configuration file
log_disconnections | on | configuration file
ssl | on | configuration file
(5 rows)
line_number | type | database | user_name | address | auth_method
14 | local | {all} | {postgres} | | peer
17 | hostssl | {comercializadora} | {app_comercializadora} | 127.0.0.1 | scram-sha-256
18 | hostssl | {comercializadora} | {app_comercializadora} | ::1 | scram-sha-256
(3 rows)
Toma la captura 06 y presiona Enter...


Windows PowerShell
== LAB-04 · Evidencia 05 · TLS · sql/04-verificacion.sql · Jesús Arturo Carrillo Avilez (20439504) ==
-- 21 La aplicación entra por TCP con TLS y comprueba el certificado del servidor
\connect "host=localhost dbname=comercializadora user=app_comercializadora sslmode=verify-full sslrootcert=/etc/ssl/certs/ssl-cert-snakeoil.pem"
Password for user app_comercializadora:
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off)
You are now connected to database "comercializadora" as user "app_comercializadora" on host "localhost" (address "::1") at port "5432".
SELECT
a.usename,
a.client_addr,
s.ssl,
s.version,
s.cipher
FROM pg_stat_ssl s
JOIN pg_stat_activity a ON a.pid = s.pid
WHERE s.pid = pg_backend_pid();
usename | client_addr | ssl | version | cipher
app_comercializadora | ::1 | t | TLSv1.3 | TLS_AES_256_GCM_SHA384
(1 row)
-- Los intentos que deben rechazarse van en un psql aparte (\!): en un script,
-- un \connect fallido cerraría esta sesión y terminaría el script.
-- La misma cuenta sin TLS: pg_hba.conf no tiene una regla para ese caso
\! psql "host=localhost dbname=comercializadora user=app_comercializadora sslmode=disable"
psql: error: connection to server at "localhost" (::1), port 5432 failed: FATAL: no pg_hba.conf entry for host "::1", user "app_comercializadora", database "comercializadora", no encryption
-- El superusuario por TCP, aunque use TLS: tampoco tiene regla
\! psql "host=localhost dbname=comercializadora user=postgres sslmode=require"
psql: error: connection to server at "localhost" (::1), port 5432 failed: FATAL: no pg_hba.conf entry for host "::1", user "postgres", database "comercializadora", SSL encryption
What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug dba-practica
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\jesus\OneDrive\Escritorio\UNIVERSIDAD\SEMESTRE 7\DBA\20439504_jesus_carrillo>